

Quick Activity 2: Canteen Group Order

A canteen sells **rice meals at ₱85 each** and **drinks at ₱25 each**. Every order receives a **10% discount on the rice-meal subtotal only**. The canteen also charges a **₱5 packaging fee per rice meal**.

Problem Analysis

Input	Process	Output
Number of rice meals, number of drinks, payment	Calculate meal subtotal, drink subtotal, discount, packaging fee, final bill, and change	Final bill and change

Decomposition (Task)

1. Read the number of rice meals.
2. Read the number of drinks.
3. Read the payment amount.
4. Calculate the meal subtotal.
5. Calculate the drink subtotal.
6. Calculate the 10% discount on the meal subtotal.
7. Calculate the packaging fee.
8. Calculate the final bill.
9. Calculate the change.
10. Display the final bill and change.

Pseudocode

START

```
READ riceMeals, drinks, payment
mealSubtotal ← riceMeals x 85
drinkSubtotal ← drinks x 25
discount ← mealSubtotal x 10%
packagingFee ← riceMeals x 5
finalBill ← mealSubtotal - discount + drinkSubtotal + packagingFee
change ← payment - finalBill
DISPLAY finalBill
DISPLAY change
```

END

Dry Run Calculations

Test Case 1

Input

- Rice Meals = 3
- Drinks = 4
- Payment = ₱500.00

Step	Variable	Calculation	Result
1	Meal Subtotal	3×85	₱255.00
2	Drink Subtotal	4×25	₱100.00
3	Discount	$255 \times 10\%$	₱25.50
4	Packaging Fee	3×5	₱15.00
5	Final Bill	$255 - 25.50 + 100 + 15$	₱344.50
6	Change	$500 - 344.50$	₱155.50

Output

Final Bill = ₱344.50

Change = ₱155.50

Test Case 2

Input

- Rice Meals = 5
- Drink = 3
- Payment = ₱600.00

Step	Variable	Calculation	Result
1	Meal Subtotal	5×85	₱425.00
2	Drink Subtotal	3×25	₱75.00
3	Discount	$425 \times 10\%$	₱42.50

4	Packaging Fee	5×5	₱25.00
5	Final Bill	$425 - 42.5 + 75 + 25$	₱482.50
6	Change	$600 - 482.50$	₱117.50

Output

Final Bill = ₱482.50

Change = ₱117.50

Explain: Why would applying the 10% discount to the entire bill produce an incorrect result? According to the instructions, the 10% discount should be applied only to the Rice Meals subtotal. If it is applied to the entire bill, the drinks and packaging fee will also receive a discount, which goes against the given rules. This will result in an incorrect final bill.